Gaussian Splatting and Neural Rendering for Dynamic Three-Dimensional Scene Reconstruction: A Comprehensive Survey

1 Introduction

The reconstruction of dynamic three-dimensional scenes represents a fundamental frontier in computer vision and graphics, bridging the gap between passive observation and interactive spatiotemporal modeling. Historically, the field has evolved from rigid multi-view stereo algorithms to flexible neural implicit representations, yet the transition from static to dynamic modeling has been fraught with computational bottlenecks and geometric ambiguities. Early approaches to dynamic view synthesis often relied on extending Neural Radiance Fields (NeRF) by incorporating time as an additional input dimension or learning deformation fields to warp a canonical space [1; 2]. While methods like [1] demonstrated the feasibility of modeling non-rigid motion from monocular video, and [3] introduced continuous scene flow functions, these implicit volumetric techniques inherently suffer from slow training convergence and prohibitive rendering costs due to dense ray marching. This computational latency has historically limited their applicability in real-time systems such as autonomous navigation or immersive telepresence, where high-frequency updates are paramount [4].

The paradigm shift toward explicit Gaussian primitives addresses these limitations by decoupling geometric representation from neural network inference, thereby enabling real-time rasterization speeds without sacrificing visual fidelity. The motivation for adopting explicit Gaussian distributions stems from the need to resolve the trade-off between the high-frequency detail capture of implicit methods and the rendering efficiency of traditional point-based graphics. Unlike implicit fields that query a neural network for every sample point along a ray, 3D Gaussian Splatting utilizes differentiable tile-based rasterization to project millions of anisotropic ellipsoids directly onto the image plane. This approach not only accelerates rendering by orders of magnitude but also provides a more intuitive interface for scene editing and physical simulation. Recent advancements have extended this explicit formulation into the temporal domain, proposing holistic 4D representations that treat spacetime as a unified volume rather than a sequence of static frames [5; 6]. By parameterizing primitives with time-dependent attributes or learning deformation fields that operate on canonical Gaussian clouds, these methods achieve real-time performance on consumer hardware while maintaining photorealistic quality in complex, non-rigid scenarios [7; 8].

However, the adoption of dynamic Gaussian Splatting introduces unique challenges distinct from static reconstruction, particularly regarding temporal consistency, motion blur modeling, and the disentanglement of static and dynamic components. The ill-posed nature of monocular dynamic reconstruction necessitates robust regularization strategies to prevent flickering artifacts and ensure geometric coherence across frames [9]. Furthermore, the integration of auxiliary priors, such as optical flow or depth estimates, has become critical for stabilizing optimization in textureless regions or under sparse view conditions [10]. This survey critically examines these architectural paradigms, ranging from canonical space warping to direct 4D primitive optimization, and evaluates their efficacy in handling diverse motion frequencies and topological changes. We further explore the integration of semantic understanding and physical constraints, which are increasingly vital for applications requiring not just visual replication but interactive scene manipulation [11; 12]. By synthesizing current methodologies and identifying persistent gaps in uncertainty quantification and large-scale scalability, this work aims to provide a rigorous foundation for the next generation of dynamic neural rendering systems.

2 Theoretical Foundations and Mathematical Formulations

2.1 Mathematical Parameterization of Spatiotemporal Primitives

The mathematical parameterization of spatiotemporal primitives constitutes the foundational layer for dynamic scene reconstruction, extending the static 3D Gaussian formulation into the fourth dimension to capture non-rigid motion and temporal evolution. In static 3D Gaussian Splatting, a scene is represented by a set of anisotropic ellipsoids defined by mean position μ, covariance Σ, opacity α, and spherical harmonic coefficients for view-dependent color. To address dynamic environments, recent paradigms diverge into two primary mathematical frameworks: continuous deformation fields that warp a canonical space, and direct 4D volumetric parameterizations that treat time as an intrinsic geometric attribute.

In the canonical space warping paradigm, primitives are anchored in a static reference frame, and their spatiotemporal attributes are derived via time-conditioned transformation functions. This approach posits that the position of a primitive at time t is given by $\mu(t) = \mu_0 + \Delta\mu(\mu_0, t)$, where $\Delta\mu$ is typically predicted by a Multi-Layer Perceptron (MLP) or a hash-grid encoded network. [8] exemplifies this by learning a deformation field to map canonical Gaussians to observation space, effectively decoupling geometric topology from motion dynamics. Similarly, [13] introduces a sparse control point mechanism where dense Gaussian motions are interpolated from compact 6-DoF transformation bases, enforcing spatial coherence through As-Rigid-As-Possible (ARAP) constraints. A critical advantage of this formulation is its memory efficiency and ability to generalize to unseen timestamps; however, it often struggles with topological changes and abrupt motions where the continuity assumption of the deformation field breaks down. To mitigate optimization instability in such continuous trajectories, [14] proposes a Dual-Domain Deformation Model, fitting attribute residuals with polynomial functions in the time domain and Fourier series in the frequency domain, thereby capturing both smooth trends and high-frequency oscillations.

Conversely, direct 4D parameterization formulations model the scene as a collection of 4D hyper-tuples, where the covariance matrix Σ_{4D} encodes correlations across space and time. [6] and [15] pioneer this approach by defining 4D anisotropic Gaussians that are sliced at specific timestamps t to yield instantaneous 3D distributions. Mathematically, this involves projecting the 4D ellipsoid onto the 3D hyperplane corresponding to the query time, naturally handling complex topology changes and object entry/exit without explicit warping functions. This method offers superior flexibility for variable-length videos and eliminates the need for separate motion networks, yet it incurs higher storage costs and requires sophisticated rendering pipelines to manage the temporal slicing and sorting of millions of 4D primitives.

Regarding appearance modeling, the extension of Spherical Harmonics (SH) into the temporal domain presents significant challenges due to the exponential growth of coefficients. While some methods opt for time-varying SH coefficients directly, others employ lightweight neural decoders or spatiotemporal feature grids to modulate view-dependent radiance, balancing fidelity against computational overhead [16]. Furthermore, the temporal opacity function $\alpha(t)$ serves as a visibility gate, allowing primitives to fade in or out to handle occlusions and transient phenomena without altering the primitive count dynamically.

Ultimately, the choice between continuous deformation and direct 4D parameterization involves a trade-off between compactness and expressiveness. While deformation-based methods like [7] excel in scenarios with persistent geometry and rigid-like motion, direct 4D approaches offer robustness against severe non-rigid deformations. Future research directions likely converge towards hybrid models that leverage the compactness of canonical spaces while incorporating the topological flexibility of 4D slicing, potentially guided by physical priors to ensure plausible intermediate states.

2.2 Differentiable Rendering Pipelines for Dynamic Volumes

Building upon the mathematical parameterizations of spatiotemporal primitives established previously, the realization of dynamic Gaussian Splatting hinges on adapting the differentiable rasterization pipeline to accommodate time-varying geometry while preserving robust gradient flow for end-to-end optimization. While static formulations rely on sorting 3D Gaussians by depth and alpha-blending their contributions, extending this to dynamic volumes necessitates a rigorous redefinition of these steps to handle the fourth dimension. In time-sliced rendering, the depth ordering of primitives must be recomputed for every timestamp t, as the trajectory of a Gaussian center $\mu(t)$ fundamentally alters its projected depth $z_i(t)$ relative to the camera. Although naive re-sorting of millions of primitives per frame incurs prohibitive computational costs, recent approaches like [15] and [14] leverage the temporal coherence of motion fields to approximate sorting orders or utilize optimized CUDA kernels that maintain differentiability during temporal slicing. This ensures that the discrete nature of rasterization does not interrupt the backpropagation required to refine the spatiotemporal trajectories defined in the preceding section.

A critical advancement in these dynamic pipelines is the integration of motion blur and exposure models, which treats frames not as instantaneous snapshots but as integrals over finite time intervals [17]. Standard splatting assumes a Dirac delta function in time, leading to artifacts when camera or object velocity is high relative to the shutter speed. To address this, [18] and [19] formalize the rendering integral by modeling the physical image formation process, where the observed color is the accumulation of radiance along the continuous trajectory of each Gaussian during the exposure window. This requires analytically integrating the time-dependent covariance $\Sigma(t)$ and opacity $\alpha(t)$, or approximating the trajectory via linear interpolation within the exposure duration. Such formulations allow the pipeline to recover sharp geometry from blurry inputs by jointly optimizing the motion parameters and the underlying scene representation, effectively disentangling motion blur from geometric uncertainty.

The gradient flow mechanisms in these dynamic pipelines face unique challenges due to the tight coupling of spatial and temporal attributes. When optimizing deformation fields or direct trajectory parameters, image-space losses must be backpropagated through the time-integrated rendering equation to update both the canonical geometry and the motion coefficients. [7] demonstrates that enforcing persistent attributes (color, size) while allowing rigid motion enables dense 6-DOF tracking purely through synthesis losses, provided the gradient signal remains stable across frames. However, long-sequence optimization often suffers from vanishing gradients or local minima where temporal inconsistencies are penalized insufficiently. To mitigate this, [14] introduces explicit supervision via optical flow, creating a differentiable bridge between pixel velocities and Gaussian dynamics. Furthermore, [8] employs annealing strategies to smooth the temporal loss landscape, facilitating convergence in monocular settings where depth-motion ambiguity is prevalent.

Comparative analysis reveals a persistent trade-off between the fidelity of motion modeling and computational efficiency within these rendering frameworks. Methods relying on implicit deformation fields, such as those discussed in [5], offer compact representations but may struggle with abrupt topological changes compared to direct 4D primitive optimization. Conversely, approaches that explicitly parameterize trajectories [20] provide superior control over high-frequency motion but require careful regularization to prevent trajectory divergence. These rendering and optimization choices directly influence the efficacy of the reconstruction, setting the stage for the advanced regularization techniques discussed in the following section, which are essential to constrain the inherent ill-posedness of spatiotemporal optimization. Future pipelines are likely to converge on hybrid schemes that combine analytic motion blur integration with hierarchical sorting structures, enabling robust reconstruction of fast-moving, non-rigid scenes from casually captured video while maintaining real-time rendering performance.

2.3 Paradigms for Encoding Temporal Dynamics and Motion Fields

The introduction of the temporal dimension into Gaussian Splatting frameworks necessitates a fundamental shift in parameterization, broadly categorized into canonical space warping, direct 4D volumetric formulations, and hybrid control-point systems. The dominant paradigm, canonical space warping, posits a static set of Gaussian primitives in a reference frame, deformed by time-conditioned functions to match observation timestamps. This approach, mathematically defined as $\mathbf{x}(t) = \mathbf{x}_0 + \Delta\mathbf{x}(\mathbf{x}_0, t)$, leverages neural networks to predict dense deformation vectors. Methods such as [8] and [7] utilize this strategy to enforce geometric consistency and persistent identity across frames, effectively disentangling appearance from motion. While computationally efficient for articulated subjects, as seen in [21], these methods often struggle with topological changes and high-frequency non-rigid deformations due to the under-constrained nature of learning inverse mappings from monocular inputs [2].

In contrast, direct 4D volumetric representations treat spacetime as a unified continuum, eliminating the need for explicit warping fields. Here, primitives are parameterized as 4D ellipsoids or voxels, where the covariance matrix encodes both spatial extent and temporal duration. [5] and [6] propose modeling the scene as a collection of 4D Gaussians, allowing for natural temporal slicing and the inherent handling of motion blur through integration over time intervals. Similarly, [22] and [23] employ decomposed tensor planes to efficiently encode spatiotemporal features, achieving significant training acceleration. Although these direct methods excel in capturing abrupt motions and complex topology changes without correspondence ambiguities, they often incur higher memory costs and may lack the explicit physical interpretability of deformation-based models.

Hybrid architectures attempt to synthesize the strengths of both paradigms by coupling dense Gaussian clouds with sparse, learnable control structures. [13] introduces a framework where sparse control points learn 6-DoF transformation bases, which are interpolated to drive the motion of dense Gaussians. This reduces the degrees of freedom, enhancing temporal coherence and enabling user-controllable motion editing. Furthermore, mesh-embedded approaches like [24] and [25] bind Gaussians to parametric surfaces or skeletons, leveraging strong geometric priors to stabilize reconstruction in sparse-view scenarios. These hybrid models offer a promising middle ground, providing the fidelity of explicit representations with the structural regularization of parametric models.

Ultimately, the choice of encoding paradigm involves a trade-off between representation capacity, computational efficiency, and temporal stability. While canonical warping offers compactness for rigid and articulated motion, direct 4D formulations provide superior flexibility for fluid and topological dynamics. Future research directions likely converge towards adaptive hybrid systems that dynamically switch between these encoding strategies based on local motion complexity, potentially integrating physics-informed constraints to ensure causal and physically plausible spatiotemporal evolution [26].

2.4 Geometric Regularization and Uncertainty Quantification

Building upon the parameterization strategies discussed previously, robust dynamic reconstruction via Gaussian Splatting necessitates rigorous regularization to resolve the inherent ill-posedness of spatiotemporal optimization. Regardless of whether a canonical warping, direct 4D, or hybrid encoding is employed, the optimization process remains susceptible to artifacts such as floating primitives, temporal flickering, and overfitting to noise, particularly in regions with sparse observations or ambiguous motion. To address geometric fidelity, recent frameworks integrate surface-aligned constraints that encourage Gaussian primitives to adhere to underlying manifolds. For instance, [27] introduces a regularization term aligning Gaussian orientations with surface normals, facilitating high-quality mesh extraction, while [28] collapses 3D volumes into 2D planar disks with depth distortion losses to ensure view-consistent geometry. In dynamic settings, [29] extends this by incorporating surface-aligned terms specifically for deformable tissues, leveraging depth-guided supervision to handle tool occlusions. Furthermore, [30] approximates surface normals at ray-Gaussian intersections, enabling direct geometry extraction through level-set identification without post-hoc fusion.

Complementing geometric constraints, ensuring temporal coherence is equally critical to prevent erratic motion and attribute fluctuations across frames. The principle of "as-rigid-as-possible" (ARAP) has emerged as a dominant strategy, where local neighborhoods of Gaussians are constrained to undergo rigid transformations, thereby preserving structural integrity during non-rigid deformation. [13] explicitly employs an ARAP loss on sparse control points to enforce spatial continuity, while [7] utilizes local-rigidity constraints to ensure persistent tracking of scene elements over time. Complementing this, [20] adopts a Dual-Domain Deformation Model, fitting attribute residuals with polynomial and Fourier series to capture long-term dependencies and filter high-frequency noise. Additionally, [31] derives an analytical velocity field from the warp network to impose Lucas-Kanade style regularization, effectively constraining both 2D optical flow and 3D trajectories for continuous space-time motion. To mitigate blur induced by rapid motion, [18] and [19] model the physical image formation process, jointly optimizing camera trajectories during exposure to recover sharp, consistent geometries from blurred inputs.

Beyond deterministic regularization, quantifying uncertainty is essential for assessing reconstruction reliability in under-constrained regions, marking a shift from static penalties to probabilistic, data-driven constraints. While early works like [32] modeled uncertainties via joint distributions of signed distance values and inlier probabilities, adapting such probabilistic frameworks to explicit Gaussian primitives remains an emerging frontier. [33] demonstrates the utility of ray-based volumetric entropy to guide view selection, a concept increasingly relevant for dynamic Gaussian systems to identify areas requiring densification or pruning. Similarly, [34] leverages monocular depth priors not merely as hard constraints but as probabilistic guides to mitigate overfitting in sparse-view scenarios. Future advancements will likely converge on hybrid variational inference frameworks that jointly optimize primitive attributes and their associated uncertainty distributions, enabling adaptive regularization strength based on local observation density and motion complexity. This evolution toward reliable, generalizable dynamic scene representation lays the groundwork for the subsequent exploration of application-specific adaptations and real-time deployment challenges.

3 Architectural Paradigms for Modeling Dynamic Motion

3.1 Canonical Space Deformation and Neural Warping Fields

The canonical space deformation paradigm represents a foundational architectural strategy for modeling dynamic scenes, wherein a static set of 3D Gaussian primitives is defined in a time-invariant reference frame and mapped to observed configurations via learned neural warping fields. This approach effectively decouples the high-frequency appearance and geometric details stored in the canonical Gaussians from the complex temporal dynamics governed by a deformation network. The core mechanism involves a coordinate transformation $\mathbf{x}(t) = \mathcal{D}(\mathbf{x}_0, t)$, where a multi-layer perceptron (MLP) or grid-based encoder predicts time-dependent displacement vectors or affine transformations for each canonical primitive $\mathbf{x}_0$. This formulation draws significant inspiration from earlier implicit methods like [1] and [35], which established the efficacy of separating static geometry from dynamic motion fields to handle non-rigid deformations from monocular inputs.

In the context of Gaussian Splatting, this paradigm has been adapted to leverage explicit geometric primitives for real-time performance. For instance, [8] employs a deformation field to warp canonical Gaussians, achieving superior rendering quality and speed compared to purely implicit volumetric methods. A critical advancement in this domain is the integration of parametric priors to constrain the ill-posed nature of unsupervised deformation learning. Methods such as [36] and [37] rig Gaussian primitives to underlying morphable models (e.g., FLAME or SMPL), utilizing the skeletal structure to guide the neural warping field. This hybridization ensures anatomical plausibility and enables controllable novel-pose synthesis, addressing the instability often observed in purely data-driven deformation networks. Similarly, [21] demonstrates that encoding Gaussians in a canonical space and transforming them via Linear Blend Skinning (LBS) allows for extremely fast training and real-time rendering, effectively mitigating the computational bottlenecks associated with per-frame optimization.

To further enhance the expressiveness and efficiency of these warping fields, recent works have introduced sparse control mechanisms and hierarchical decomposition. [13] proposes decomposing motion into sparse control points that drive dense Gaussian deformations, significantly reducing learning complexity while enabling user-controlled motion editing. This contrasts with dense per-Gaussian deformation fields, offering a more compact representation of articulated motion. Furthermore, [38] argues against generic coordinate-based deformation functions, proposing instead that deformation should be conditioned on per-Gaussian embeddings to better capture local motion characteristics and prevent blurring in dynamic regions. Despite these advances, challenges remain in handling topological changes and ensuring long-term temporal consistency without overfitting to noise. Future directions likely involve tighter coupling with physical simulation constraints and the development of adaptive warping fields that can dynamically adjust their capacity based on local motion complexity, bridging the gap between data-driven flexibility and physical plausibility.

3.2 Direct Spatiotemporal Primitive Parameterization

Complementing the canonical space deformation paradigm, which decouples static geometry from dynamic motion fields, direct spatiotemporal primitive parameterization treats time as an intrinsic dimension of the geometric representation itself. Rather than relying on implicit neural warping functions to map a static reference to observation space, this approach redefines the Gaussian primitive from a static 3D ellipsoid to a dynamic entity evolving continuously or discretely over the temporal domain. This fundamental shift bypasses the computational overhead and optimization instability often associated with learning complex warp functions, offering a more explicit mechanism for modeling dynamics. A primary manifestation of this strategy is the extension of 3D Gaussians into 4D spacetime ellipsoids, where the covariance matrix encapsulates both spatial extent and temporal duration. As demonstrated in [5], representing the scene as a collection of 4D primitives allows for the natural modeling of motion blur and abrupt topological changes through temporal slicing, effectively approximating the underlying spatiotemporal volume without explicit motion vectors. Similarly, [6] utilizes anisotropic 4D XYZT Gaussians with 4D spherindrical harmonics to capture view-dependent and time-evolved appearance, achieving superior efficiency in handling complex dynamics compared to frame-wise optimization.

Beyond holistic 4D formulations, trajectory-based motion modeling offers a compact alternative by parameterizing the temporal evolution of Gaussian centers using learnable motion bases. Instead of storing per-frame attributes, methods like [20] employ a Dual-Domain Deformation Model that captures time-dependent residuals via polynomial fitting in the time domain and Fourier series in the frequency domain. This explicit deformation modeling eliminates the need for separate implicit neural fields, enabling ultra-fast training and rendering comparable to static 3D Gaussian Splatting. Furthermore, [16] enhances this concept by introducing temporal opacity and parametric motion, allowing primitives to capture static, dynamic, and transient content within a unified framework while replacing spherical harmonics with splatted neural features for efficient appearance encoding. To further reduce the degrees of freedom and enforce spatial coherence, clustering strategies group neighboring Gaussians into “superpoints” or control points that share rigid or affine motion parameters. [13] exemplifies this by decomposing motion and appearance into sparse control points and dense Gaussians, where a deformation MLP predicts 6-DoF transformations for control points that are locally interpolated to drive the motion field, significantly enhancing temporal consistency and enabling user-controlled motion editing.

The direct parameterization strategy offers distinct advantages in rendering speed and memory efficiency, particularly for long sequences where canonical mappings may suffer from error accumulation or excessive network capacity requirements. However, this approach faces challenges in generalizing to unseen timestamps outside the training range and managing the substantial memory footprint of high-resolution 4D feature grids. Recent works address these limitations through adaptive densification and pruning mechanisms, such as the deformation-aware pruning in [39], which gauges the impact of each Gaussian on deformation to minimize redundancy. Additionally, [40] introduces a Neural Transformation Cache to model translations and rotations compactly, facilitating on-the-fly reconstruction for streaming applications. While direct spatiotemporal parameterization excels in capturing high-frequency details and ensuring real-time performance, it often struggles with topological consistency and anatomical plausibility in sparse-view or highly articulated scenarios without additional constraints. Consequently, future research increasingly points toward hybrid schemes that integrate the structural priors of parametric models with the flexibility of explicit 4D primitives, bridging the gap between explicit flexibility and scalable modular decomposition to address the inherent ill-posedness of dynamic reconstruction.

3.3 Hybrid Representations with Parametric and Physical Priors

Hybrid representations integrating explicit Gaussian clouds with parametric and physical priors address the inherent ill-posedness of dynamic reconstruction by constraining the optimization manifold with structured knowledge. While pure deformation fields offer flexibility, they often struggle with topological consistency and anatomical plausibility in sparse-view or highly articulated scenarios. Consequently, recent architectures increasingly couple the high-frequency radiance fidelity of 3D Gaussians with the geometric rigor of meshes, skeletal rigs, and physics engines.

A dominant strategy involves anchoring Gaussian primitives to deformable mesh surfaces, leveraging the mesh's topology to enforce spatial coherence while retaining the splatting pipeline's rendering efficiency. In [24], Gaussians are defined via barycentric coordinates and displacement vectors on triangle meshes, effectively disentangling low-frequency motion (handled by the mesh) from high-frequency appearance details. This mesh-embedded approach facilitates compatibility with standard animation techniques like blend shapes and skeletal animation, overcoming the limitations of MLP-based skinning fields. Similarly, [41] parameterizes Gaussian components directly by mesh vertices, enabling automatic adjustments in position, scale, and rotation during animation, thus providing a real-time editable framework. For head avatars, [36] rigs 3D Gaussians to a parametric morphable face model, optimizing explicit displacement offsets to achieve photorealistic expression control. These surface-constrained methods significantly reduce the degrees of freedom required for optimization, mitigating artifacts such as floating points and ensuring that the reconstructed geometry remains manifold-consistent even under large deformations.

Parallel to mesh-based approaches, skeletal priors provide strong regularization for articulated human motion. Methods like [42] and [21] utilize the SMPL body model to initialize Gaussian distributions and employ Linear Blend Skinning (LBS) to transform primitives from a canonical space to observation space. By jointly optimizing skinning weights alongside Gaussian attributes, these frameworks resolve ambiguities in cloth and hair dynamics that static templates cannot capture. [37] further enhances this by introducing as-isometric-as-possible regularizations on Gaussian covariance matrices, stabilizing learning under highly articulated unseen poses. The integration of skeletal structures not only accelerates convergence—reducing training times from hours to minutes—but also enables novel-pose synthesis, a capability often absent in purely data-driven deformation models.

Beyond kinematic constraints, emerging paradigms incorporate physical simulation engines to enforce dynamic plausibility. [12] demonstrates the integration of Position-Based Dynamics (PBD) with Gaussian Splatting, where Gaussian normals are aligned with surface normals to refine fluid and solid interactions. This coupling allows for the realistic synthesis of surface highlights on dynamic fluids and coherent object interactions. Furthermore, [26] proposes a dual Gaussian-particle representation where particles govern physical states and Gaussians handle visual rendering. This system generates "visual forces" by comparing predicted and observed images, enabling online correction of physical states in real-time. Such physics-informed constraints are critical for robotics applications, as seen in [43], which distills vision-language features to assign material properties for particle-based simulations grounded in natural language.

Despite these advancements, challenges remain in balancing the rigidity of parametric priors with the flexibility needed for non-rigid details. Over-reliance on templates can suppress fine-grained motion, while loose coupling may fail to prevent interpenetration. Future research directions point toward adaptive hybridization, where the strength of parametric constraints varies dynamically based on observation confidence, and tighter integration with differentiable physics solvers to enable fully interactive, physically consistent world models.

3.4 Adaptive Scene Decomposition and Modular Architectures

Building on the constrained flexibility of hybrid representations, adaptive scene decomposition and modular architectures address the computational intractability and optimization instability inherent in modeling complex, unbounded spatiotemporal volumes as monolithic entities. Where previous approaches relied on global parametric priors to regularize deformation, this paradigm disentangles scenes into independent, manageable components—ranging from semantic objects to temporal windows—to enable scalable resource allocation and robust reconstruction under varying motion frequencies. A primary strategy involves object-centric decomposition, where attention-based mechanisms isolate dynamic entities into distinct Gaussian slots. For instance, [11] augments primitives with identity encodings supervised by 2D masks, facilitating fine-grained segmentation and editing without expensive 3D labels. Similarly, [44] distills semantic features from foundation models into Gaussian attributes, enabling open-vocabulary scene understanding and modular manipulation. This object-centric modularity is further advanced in robotics applications, where [45] leverages editable Gaussian representations to maintain a "digital twin" of evolving environments, allowing for multi-stage manipulation tasks that require persistent object tracking and scene updating.

Complementing spatial separation, temporal decomposition offers a potent solution for handling long-duration sequences and significant topological changes that challenge continuous deformation fields. Rather than optimizing a single global trajectory, methods like [46] decompose the 4D spatiotemporal space based on temporal characteristics, assigning static, deforming, and emerging regions to separate neural fields regularized by distinct priors. This approach effectively mitigates the catastrophic forgetting often observed in continuous training streams. Complementing this, [47] adopts an incremental learning paradigm, training per-frame model differences to complement a base model, thereby achieving efficient on-the-fly adaptation with minimal storage overhead. For scenes exhibiting abrupt motions or variable-length dynamics, [15] treats spacetime as a unified volume but employs temporal slicing to compose dynamic 3D Gaussians, effectively decoupling the rendering complexity from the duration of the event. Furthermore, [48] introduces 4D motion scaffolds to compactly encode underlying deformations, disentangling geometry from the deformation field to stabilize optimization in casual monocular settings.

Layered representations provide another architectural avenue for resolving occlusion ambiguities and separating foreground dynamics from static backgrounds, a critical step for scaling to large environments. [49] factorizes large-scale urban scenes into three separate hash table data structures encoding static, dynamic, and far-field radiance fields, enabling unsupervised 3D instance segmentation and scaling to millions of frames without requiring explicit bounding box supervision. This static-dynamic disentanglement is crucial for optimizing computational resources, as it allows high-frequency updates to be applied solely to moving components while maintaining a stable background representation. The integration of such modular designs with the physical priors discussed previously creates a powerful synergy; as seen in [43], this combination allows for the synthesis of physics-based dynamics where material properties are assigned automatically via text queries, bridging the gap between visual reconstruction and interactive simulation.

Despite these advancements, challenges remain in seamlessly integrating these modular components without introducing boundary artifacts or temporal flickering. Current approaches often struggle with the consistent propagation of appearance and geometry across decomposed boundaries, particularly when objects interact or merge. Future research directions should focus on developing unified loss functions that enforce cross-module consistency while preserving the computational benefits of decomposition. Additionally, the integration of neuro-symbolic reasoning could enable higher-level scene understanding, allowing modular architectures to dynamically reconfigure their decomposition strategies based on semantic context rather than fixed heuristics. As the field progresses, the synergy between adaptive decomposition and generative priors will likely define the next generation of scalable, interactive dynamic scene representations.

4 Optimization Strategies and Geometric Regularization

4.1 Adaptive Primitive Management and Topological Control

Adaptive primitive management constitutes the cornerstone of efficient dynamic Gaussian Splatting, addressing the fundamental tension between representing complex, time-varying geometry and maintaining computational tractability. In static scenarios, density control relies primarily on view-space positional gradients to split or clone primitives; however, dynamic environments introduce temporal dimensions where motion velocity, deformation magnitude, and topological evolution dictate primitive lifecycles. A naive extension of static densification strategies often leads to over-reconstruction in high-motion regions or temporal flickering due to inconsistent primitive birth and death rates across frames. To mitigate this, recent advancements have shifted toward motion-aware criteria that couple spatial error metrics with temporal coherence constraints. For instance, [50] leverages optical flow cues to guide densification, ensuring that primitives are allocated proportionally to motion complexity rather than mere photometric error, thereby reducing redundancy in static background regions while preserving detail in rapidly deforming foregrounds.

The challenge of topological control in dynamic scenes is further compounded by the need to prevent "gradient collision," a phenomenon where large primitives in over-reconstructed regions fail to split due to conflicting gradient signals, leading to blurry artifacts. [51] addresses this by introducing homodirectional view-space positional gradients as a more robust splitting criterion, which effectively recovers fine details in intricate dynamic scenes. Furthermore, maintaining temporal consistency requires mechanisms that enforce persistent identity for primitives over time. [7] demonstrates that enforcing persistent color, opacity, and size attributes, regularized by local rigidity constraints, allows the system to naturally track dense scene elements without explicit correspondence, effectively stabilizing the topology against noisy optimization trajectories. Conversely, methods like [13] decouple motion from appearance by utilizing sparse control points to drive dense Gaussian transformations, significantly reducing the degrees of freedom required for topology management and enabling coherent motion editing.

Memory efficiency remains a critical bottleneck, particularly when modeling long sequences where primitive counts can explode. Innovative compression strategies have emerged to regulate primitive distribution through quantization and pruning. [52] and [53] propose vector quantization and network pruning techniques that identify and remove insignificant Gaussians, achieving substantial storage reduction without compromising visual fidelity. In the specific context of dynamic surgical scenes, [39] introduces deformation-aware pruning, which gauges the impact of each Gaussian on the deformation field to eliminate redundancy, while [54] employs a flexible deformation modeling scheme to accelerate reconstruction by optimizing individual Gaussian dynamics. Additionally, hierarchical approaches such as [55] and [56] adapt Level-of-Detail (LOD) structures to the temporal domain, dynamically selecting primitive resolutions based on viewing distance and motion speed to ensure consistent rendering performance.

Despite these advances, achieving a unified framework that seamlessly handles abrupt topological changes—such as object splitting or merging—remains an open challenge. Current methods often rely on heuristic thresholds for pruning and splitting, which may fail under extreme non-rigid deformations. Future research directions should explore learnable lifecycle policies driven by uncertainty estimation, allowing the system to probabilistically activate or deactivate primitives based on observational confidence, thereby creating a more robust and adaptive representation for unconstrained dynamic environments.

4.2 Integration of Auxiliary Geometric and Semantic Priors

While the integration of auxiliary priors discussed in this section provides essential geometric and semantic constraints, their role is particularly critical in stabilizing the inherently ill-posed optimization of dynamic 3D Gaussian Splatting (3DGS). This stabilization serves as a necessary precursor to the adaptive primitive management strategies detailed in the subsequent section, as reliable priors prevent the chaotic birth and death of primitives caused by monocular ambiguities, textureless surfaces, or complex non-rigid motions. Recent paradigms increasingly leverage dense depth and optical flow estimates derived from pre-trained foundation models and multi-view stereo algorithms to anchor the spatiotemporal trajectories of Gaussian primitives. In sparse-view or monocular settings, where photometric loss alone fails to constrain depth, methods incorporate monocular depth estimators as soft regularization terms. For instance, [34] aligns sparse Structure-from-Motion points with dense monocular depth maps to guide Gaussian positioning, effectively suppressing floating artifacts. Similarly, [42] introduces global-local depth normalization to restore accurate geometry under coarse supervision, while [57] utilizes depth-based initialization coupled with flow-based losses to enforce multi-view consistency in extremely sparse input scenarios.

Beyond depth, optical flow serves as a critical supervisory signal for modeling dynamic deformations, directly addressing the temporal inconsistencies that complicate primitive lifecycle management. The [14] framework explicitly connects 3D Gaussian dynamics to pixel velocities, splatting Gaussian motion into image space to derive a differentiable "Gaussian flow" that directly regularizes temporal coherence. This approach resolves color drifting and ensures that primitive motions align with observed 2D displacements. Complementing this, [50] mines uncertainty-aware motion cues from optical flow to enhance deformation fields, demonstrating that integrating 2D motion priors significantly reduces model redundancy and improves rendering fidelity in dynamic regions. Furthermore, [58] extends this by utilizing only 2D depth and flow priors to lift monocular videos into explicit 4D representations, simultaneously optimizing camera poses and scene dynamics without external tracking systems, thereby laying the groundwork for the pose-free reconstruction techniques explored later.

To address the challenge of reconstructing coherent surfaces from discrete, unorganized Gaussians, geometric regularization via surface normals and Signed Distance Functions (SDF) has become indispensable. Standard 3DGS often struggles to align primitives with underlying manifolds, leading to noisy geometry that hinders effective densification. [42] introduces a regularization term that encourages Gaussians to align with surface normals, facilitating high-quality mesh extraction via Poisson reconstruction. Building on this, [59] refines point clouds by iteratively updating normal priors predicted by neural implicit models, creating a symbiotic optimization loop between explicit Gaussians and implicit surfaces. In indoor environments characterized by large textureless planes, [60] and [61] integrate estimated normal maps as adaptive priors, enforcing local smoothness while preserving high-frequency details in textured regions. Additionally, [62] explicitly extracts 3D geometry features to constrain deformation learning, ensuring that non-rigid motions remain geometrically coherent rather than merely photometrically consistent.

Semantic priors further enhance reconstruction by disentangling static backgrounds from dynamic foregrounds, a distinction that is vital for the motion-aware splitting and pruning mechanisms discussed previously. [11] augments primitives with identity encodings supervised by 2D segmentation masks (e.g., from SAM), allowing for fine-grained scene decomposition and editing. This semantic awareness is crucial for dynamic scenes; [42] jointly optimizes geometry, appearance, and semantics in urban environments, regularizing moving object poses via physical constraints to handle noisy detection inputs. Similarly, [63] distills vision-language embeddings from foundation models into 3D Gaussians, achieving holistic scene understanding that supports open-vocabulary queries. By integrating these diverse auxiliary signals—depth, flow, normals, and semantics—modern dynamic 3DGS frameworks transcend the limitations of pure image-based optimization, achieving robust, high-fidelity reconstructions even in under-constrained and highly dynamic environments. Future research directions point towards tighter coupling with generative priors to hallucinate occluded regions and the development of unified loss functions that balance conflicting geometric and semantic constraints adaptively, bridging the gap toward the fully unconstrained, joint optimization frameworks examined in the following section.

4.3 Robust Optimization Under Uncertain Conditions

Dynamic scene reconstruction from casual, in-the-wild video captures presents a severely ill-posed optimization problem, characterized by unknown camera trajectories, motion blur, and sparse observational data. Traditional pipelines relying on accurate Structure-from-Motion (SfM) initialization often fail under these conditions, necessitating end-to-end frameworks that jointly estimate scene geometry and camera poses. Recent advancements in Gaussian Splatting (GS) have pivoted toward bundle-adjusting strategies that resolve large pose misalignments while optimizing explicit primitives. For instance, [18] and [64] model the physical image formation process of motion-blurred frames, jointly recovering camera motion trajectories during exposure times and Gaussian parameters to prevent geometric smearing. Similarly, [60] integrates visual-inertial odometry priors to compensate for rolling shutter distortions and non-static camera poses within single exposure intervals, demonstrating that explicit geometric representations offer superior convergence basins compared to implicit volumetric fields when handling severe motion artifacts.

The challenge of unknown poses is further addressed by methods that eliminate the dependency on pre-computed extrinsics entirely. [65] leverages the continuity of video streams to progressively grow Gaussian sets sequentially, optimizing poses and geometry simultaneously without SfM preprocessing. This approach is complemented by [66], which unifies dense stereo priors from foundation models like DUSt3R with 3D-GS to achieve pose-free reconstruction in under a minute, significantly reducing Absolute Trajectory Error in sparse-view scenarios. Furthermore, [58] extends this capability to monocular videos by lifting 2D depth and optical flow priors into a 4D explicit representation, enabling the recovery of dynamic worlds and camera poses from a single uncalibrated sequence. These methods collectively illustrate a paradigm shift from sequential preprocessing to coupled optimization, where geometric consistency constraints regularize the pose estimation process.

In scenarios with sparse input views, the risk of overfitting and geometric degeneration is acute. To mitigate this, [57] introduces structured regularization through implicit convolutional decoders and total variation losses, constraining Gaussian positions to maintain 2D coherence and preventing the formation of needle-like artifacts. Complementary approaches like [42] employ global-local depth normalization, utilizing coarse monocular depth supervision to anchor Gaussian positions while preserving high-frequency appearance details. The integration of generative priors has also emerged as a potent strategy for handling under-constrained regions. [67] and [68] distill view-conditioned diffusion models to regularize Novel View Synthesis, effectively hallucinating plausible geometry and texture in unobserved areas while maintaining 3D consistency. Additionally, [69] addresses photometric variations and transient occluders in unconstrained photo collections by decomposing appearance into intrinsic and dynamic components, ensuring robust reconstruction despite varying illumination and weather conditions.

Despite these advances, trade-offs remain between computational efficiency and the robustness of joint optimization. While methods like [70] established the theoretical groundwork for coarse-to-fine registration in neural fields, their translation to explicit Gaussian primitives requires careful management of primitive density to avoid local minima. Future research directions should focus on enhancing the uncertainty quantification within these joint optimization frameworks, potentially integrating probabilistic modeling similar to [71] to handle multimodal pose distributions in highly symmetric or textureless environments. Ultimately, the convergence of physical image formation models, dense stereo priors, and generative regularization is defining a new standard for robust dynamic reconstruction in unconstrained settings.

4.4 Temporal Regularization and Dynamic Consistency

Building upon the robust pose estimation and geometric regularization strategies established for unconstrained inputs, the subsequent challenge lies in ensuring temporal consistency within dynamic Gaussian Splatting. This step is paramount for mitigating flickering artifacts and preserving physical plausibility during novel view synthesis. Unlike static reconstruction, where geometric fidelity is the primary objective, dynamic settings require rigorous regularization of the temporal evolution of primitive attributes to enforce smoothness, causality, and rigidity where appropriate. A dominant strategy involves imposing smoothness constraints on the deformation fields that warp canonical Gaussians to observation space. Methods such as [13] employ an As-Rigid-As-Possible (ARAP) loss to penalize deviations from local rigidity, ensuring that neighborhoods of control points maintain their relative structural integrity during non-rigid motion. Similarly, [35] introduces an elastic regularization term derived from physical simulation principles to prevent the deformation network from collapsing into local minima, thereby stabilizing the optimization of non-rigidly deforming scenes captured casually. These spatial-temporal priors are critical for disentangling true geometric motion from appearance fluctuations, a common failure mode in under-constrained monocular settings.

Beyond spatial coherence, recent advances have leveraged frequency-domain analysis to regulate temporal dynamics, addressing limitations inherent in purely time-domain optimization. [20] proposes a Dual-Domain Deformation Model that explicitly fits the time-dependent residuals of Gaussian attributes using Fourier series, effectively capturing periodic motions while filtering out high-frequency noise that manifests as jitter. This spectral approach complements spatial regularization by addressing long-term dependencies that point-wise smoothness terms often miss. Furthermore, [72] extends this concept to endoscopic scenes by introducing Temporal High-Frequency Emphasis Reconstruction, which utilizes optical flow priors to specifically target and enhance motion-intensive regions, ensuring that rapid deformations are reconstructed without blurring. The integration of such frequency-aware losses marks a significant shift toward hybrid spectral-spatial frameworks that better preserve dynamic detail.

Causal consistency and bidirectional warping represent another frontier in temporal regularization, aiming to resolve ambiguities inherent in forward warping. To this end, [73] advocates for estimating smooth forward flow fields rather than discontinuous backward maps, facilitating more robust motion modeling. Complementing this, [74] leverages cycle-consistency losses across casual videos to self-supervise dense correspondences, ensuring that forward and backward deformations are invertible and temporally stable. This is particularly effective when combined with neural blend skinning models, which provide a structured prior for articulated motion. Additionally, [48] utilizes a 4D Motion Scaffold to compactly encode underlying motions, disentangling geometry from deformation and allowing for seamless camera pose refinement alongside dynamic rendering.

However, the efficacy of these methods relies heavily on the assumption that motion trajectories are continuous and differentiable, an assumption challenged by abrupt topological changes.

Despite these advances, trade-offs remain between computational efficiency and the strictness of temporal constraints. While [13] and [20] achieve real-time performance by decoupling motion bases from dense appearance, methods relying on heavy cycle-consistency checks [74] or complex spectral decompositions may incur higher training overheads. Future research directions point towards adaptive regularization schemes that dynamically adjust smoothness weights based on local motion uncertainty, potentially integrating differentiable physics engines [75] to enforce conservation laws directly on Gaussian trajectories. Such physics-informed temporal regularization promises to further enhance the realism of dynamic reconstructions, moving beyond visual coherence to true physical simulatability.

5 Advanced Capabilities and Interactive Scene Manipulation

5.1 Inverse Rendering and Relightable Dynamic Factorization

Inverse rendering within dynamic Gaussian Splatting (3DGS) frameworks represents a pivotal shift from passive view synthesis to active scene manipulation, necessitating the rigorous disentanglement of geometry, intrinsic materials, and extrinsic illumination. While standard 3DGS excels in novel view synthesis, its native reliance on view-dependent spherical harmonics entangles lighting with appearance, precluding relighting capabilities. Recent advancements address this by augmenting Gaussian primitives with physical shading models. For instance, [76] introduces a simplified shading function coupled with a novel normal estimation framework derived from the shortest axis of anisotropic covariances, enabling high-fidelity rendering of reflective surfaces without sacrificing real-time performance. Similarly, [77] extends this by incorporating depth-derivation regularization for normal consistency and a baking-based occlusion model to approximate indirect lighting, effectively bridging the gap between forward-mapping rasterization and inverse rendering requirements.

The challenge of estimating reliable surface normals from discrete, unstructured Gaussians remains central to physically based shading. [78] proposes a comprehensive solution by associating each primitive with explicit BRDF parameters and decomposing incident light into global and local components. Crucially, this method employs a point-based ray-tracing approach via bounding volume hierarchies to bake visibility, allowing for accurate shadow casting and real-time relighting. This contrasts with implicit methods like [79] or [80], which, while robust for static objects, struggle with the computational demands of dynamic scenes. In the dynamic domain, [71] decomposes space-time varying human geometry into neural fields of normal, occlusion, and reflectance maps, integrating them into a reflectance-aware rendering pipeline that supports free-viewpoint relighting from monocular video. Furthermore, [81] addresses the specific failure modes of dynamic specular objects by reformulating the radiance field to condition on observation-space surface orientations, ensuring that reflected colors remain consistent despite non-rigid deformations.

A critical trade-off exists between the expressiveness of the material model and optimization stability. Approaches utilizing microfacet BRDFs, as seen in [82] and adapted for Gaussians in [83] via anisotropic spherical Gaussians (ASG), offer superior modeling of high-frequency specularities compared to standard spherical harmonics. However, these methods require precise normal alignment, often necessitating strong geometric priors or regularization terms such as those proposed in [84], which leverages depth and normal cues to enforce local smoothness. The integration of global illumination remains an open frontier; while [85] simulates ray tracing within the Gaussian framework to reconstruct indirect lighting, many dynamic methods currently rely on simplified lighting assumptions to maintain tractability.

Looking forward, the convergence of explicit Gaussian representations with differentiable physics and generative priors promises to resolve remaining ambiguities in material-lighting decomposition. The ability to edit materials and lighting in dynamic scenes, as demonstrated by [86] for neural characters, suggests a trajectory toward fully interactive world models. Future research must focus on unifying these static inverse rendering successes with robust temporal consistency constraints, ensuring that material properties remain invariant across time while accommodating complex topological changes and motion blur, thereby enabling true photorealistic editing of 4D scenes.

5.2 Semantic-Aware Scene Understanding and Granular Editing

While the preceding section established the physical foundations for relighting and material editing through inverse rendering, the integration of semantic intelligence with explicit Gaussian primitives marks a complementary paradigm shift from passive synthesis to active, object-level scene manipulation. Unlike implicit neural fields where semantics are often entangled within continuous volumetric functions, the discrete nature of 3D Gaussian Splatting (3DGS) facilitates the direct association of high-dimensional feature vectors with individual primitives. This architectural advantage enables open-vocabulary scene understanding by distilling pre-trained 2D foundation model embeddings, such as CLIP or SAM, directly into the Gaussian attribute space. For instance, [87] and [88] demonstrate efficient strategies for projecting 2D semantic features onto 3D Gaussians, employing compact semantic components and self-training consistency losses to resolve view-dependent ambiguities without sacrificing real-time rendering performance. Similarly, [44] addresses the resolution disparities between RGB and feature maps, enabling precise point and bounding-box prompting for radiance field manipulation. These methods collectively establish a robust framework for "Segment Anything" in 3D, where [11] leverages identity encodings supervised by 2D masks to cluster Gaussians into coherent object instances, effectively disentangling dynamic foregrounds from static backgrounds.

Building upon this semantic grounding, granular editing capabilities have emerged as a critical application, allowing for language-guided or mask-based modification of scene content. The explicit representation permits the selective optimization of Gaussian subsets, enabling operations such as object removal, colorization, and structural deformation with unprecedented efficiency. [89] introduces hierarchical Gaussian splatting combined with semantic tracing to achieve swift, controllable editing, while [90] utilizes score distillation sampling (SDS) from diffusion models to alter object styles and appearances based on textual instructions. A significant challenge in this domain is maintaining multi-view consistency during editing; [91] and [92] address this by enforcing geometric and attention-based consistency across latent codes, ensuring that edits applied to reference views propagate coherently throughout the 3D structure. Furthermore, [93] exploits DINO feature correspondence to transfer texture and style changes from single reference images to the entire 3D model, demonstrating the versatility of feature-matched editing pipelines.

The synergy between semantics and dynamics further extends these capabilities to articulated and non-rigid scenarios, bridging the gap toward the generative completion tasks discussed in the subsequent section. [13] decomposes motion and appearance using sparse control points, allowing for user-controlled motion retargeting of specific semantic groups while preserving high-fidelity appearance. In complex urban or robotic environments, [94] and [95] integrate semantic propagation with dynamic Gaussian frameworks to support holistic scene understanding and multi-task manipulation, respectively. Despite these advances, trade-offs remain between the dimensionality of distilled features and computational overhead. While [63] incorporates vision-language embeddings for holistic understanding, the memory footprint of high-dimensional feature vectors necessitates efficient compression strategies, as explored in [96] through lightweight encoder-decoder architectures. Future research must address the temporal stability of semantic labels in highly dynamic scenes and develop unified frameworks that seamlessly couple semantic reasoning with physical simulation constraints. By establishing these semantically aware, object-level manipulation capabilities, the field creates the necessary grounding for the diffusion-driven generative priors and sparse-view completion strategies detailed in the following section.

5.3 Generative Completion and View Extrapolation via Diffusion Priors

The integration of diffusion priors with 3D Gaussian Splatting (3DGS) represents a paradigm shift in addressing the ill-posed nature of sparse-view reconstruction and view extrapolation. While explicit Gaussian primitives offer real-time rendering fidelity, they inherently lack the generative capacity to hallucinate plausible geometry and texture in unobserved regions. Recent methodologies bridge this gap by distilling knowledge from large-scale 2D and 3D diffusion models into the explicit Gaussian representation, effectively regularizing the optimization landscape where data constraints are absent.

A primary strategy involves leveraging multi-view consistent diffusion to synthesize novel views that guide Gaussian optimization. Approaches such as [97] simulate a capture process by generating highly consistent novel views from sparse inputs, which are subsequently fused into a robust 3DGS representation. Similarly, [98] employs a pose-free architecture to synthesize dense, high-resolution views from single or sparse images, bypassing the need for explicit camera parameters during generation. These generated views serve as pseudo-ground truth, enabling 3DGS to converge on geometrically coherent structures that traditional photometric losses alone cannot recover. Furthermore, [99] unifies single- and multi-image reconstruction by learning implicit 3D representations coupled with camera positional encodings, allowing for scalable view synthesis that maintains consistency across arbitrary camera transformations.

Beyond view synthesis, diffusion priors are critical for completing occluded regions and refining geometric details. [67] introduces a framework where a diffusion prior regularizes a NeRF-based pipeline at novel camera poses, a concept increasingly adapted for Gaussian splatting to preserve appearance in observed regions while synthesizing realistic geometry in under-constrained areas. In the domain of dynamic scenes, [100] utilizes diffusion priors to construct animatable Gaussian splats from monocular videos, overcoming the limitations of limited view coverage through rigid regularization and generative completion. For static scenes, [101] proposes a “Dreaming and Alignment” pipeline, using point clouds as geometric guidelines for inpainting via generative models, which are then lifted and aligned into a unified 3DGS scene. This approach effectively addresses domain gaps by leveraging internet-scale priors rather than restricted 3D scan datasets.

To ensure multi-view consistency during editing and completion, advanced attention mechanisms are employed. [91] introduces depth-conditioned editing and attention-based latent code alignment to enforce geometric and appearance consistency across multi-view images during text-driven editing, mitigating the mode collapse often seen in iterative 2D editing workflows. Complementing this, [102] incorporates cross-attention and editing consistency modules to resolve discrepancies in guidance images, ensuring that the updated Gaussian primitives maintain structural integrity across viewpoints. Additionally, [103] demonstrates that distilling view-conditioned latent diffusion models allows for the recovery of plausible 3D representations where geometric consistency and generative inference act complementarily, achieving superior performance in sparse-view novel view synthesis.

Despite these advances, challenges remain in balancing the stochastic nature of diffusion models with the deterministic requirements of 3D geometry. Methods like [104] highlight the instability in sparse-view settings, proposing co-regularization strategies to suppress inaccurate reconstructions that may arise from overfitting to noisy generative priors. Future research directions point towards tighter coupling of 3D-aware diffusion architectures, such as [105], which structures Gaussians via optimal transport to enable standard 3D U-Nets as backbones, thereby facilitating more native and efficient generative modeling directly within the Gaussian domain. This evolution promises to unlock fully generative 3DGS pipelines capable of creating complex, dynamic scenes from minimal semantic or visual prompts.

5.4 Physics-Informed Simulation and Interactive World Models

Building on the generative capabilities of diffusion-enhanced 3DGS discussed previously, the integration of differentiable rendering with physics simulation marks a pivotal evolution in dynamic scene reconstruction. This transition shifts Gaussian Splatting from passive visual synthesis toward active, interactive world models, addressing the fundamental limitations of purely data-driven approaches that often lack physical plausibility and struggle with counterfactual reasoning. By endowing explicit Gaussian primitives with physical attributes, recent methodologies enable robust collision handling, dynamic interaction, and predictive simulation essential for robotics and virtual reality.

A primary strategy involves coupling Gaussian representations with differentiable physics engines to simulate rigid and soft-body dynamics. [12] demonstrates the feasibility of integrating Position-Based Dynamics (PBD) with 3D Gaussian Splatting, where Gaussian normals are aligned with surface constraints to eliminate rotational artifacts and realistically render fluid interactions. Similarly, [43] bridges the gap between semantics and physics by distilling vision-language features into Gaussians, allowing material properties to be assigned via text queries and synthesized into particle-based simulations. This unification facilitates semi-automatic scene decomposition and the generation of physically grounded dynamics from static observations. Furthermore, [106] advances this interaction within Virtual Reality by employing a two-level embedding strategy that supports real-time deformable body simulations, ensuring highly realistic dynamic responses to user manipulation.

Beyond offline simulation, emerging frameworks establish closed-loop systems for online state estimation and correction, which are critical for robotic autonomy. [26] proposes a dual Gaussian-Particle representation where particles model geometry for predictive simulation while attached Gaussians capture visual appearance. By comparing predicted and observed images, the system generates "visual forces" to correct particle positions in real-time, effectively synchronizing the physical model with reality despite dynamic changes. This approach contrasts with traditional SLAM methods like [4], which primarily focus on masking dynamic objects rather than modeling their physical evolution. In low-light or unstructured environments, [107] extends these capabilities by learning neural illumination models to reconstruct relightable scenes, thereby supporting robotic exploration where lighting conditions are non-stationary.

For complex manipulation tasks, the explicit nature of Gaussians allows for the integration of semantic affordances directly into the physical model. [45] leverages editable Gaussian Splatting to create a "digital twin" of the environment, utilizing semantic masking and infilling to visualize object motions resulting from robot interactions. This enables multi-stage, open-vocabulary manipulation by maintaining a consistent scene representation that evolves with the task. Additionally, [95] formulates a Gaussian world model that mines scene dynamics to predict optimal robot actions, outperforming conventional methods by incorporating spatiotemporal context into the decision-making loop.

Despite these advancements, challenges remain in balancing computational efficiency with physical fidelity. While methods like [12] and [106] achieve real-time performance, scaling these simulations to large-scale, highly deformable environments without sacrificing visual quality remains an open problem. Future research directions point toward neuro-symbolic integration, where the continuous differentiability of Gaussian Splatting is combined with explicit reasoning engines to handle complex causal relationships and long-horizon planning. The synthesis of generative priors with physics-informed constraints, as hinted at by the success of [43] in material assignment, suggests a trajectory toward world models that are not only visually photorealistic but also causally consistent and universally interactive.

6 Systems, Applications, and Performance Evaluation

6.1 Real-Time Robotic Perception and Navigation Systems

The integration of dynamic Gaussian Splatting into robotic perception pipelines marks a paradigm shift from implicit neural fields to explicit, differentiable volumetric representations capable of supporting real-time simultaneous localization and mapping (SLAM) in non-static environments. Unlike traditional mesh-based or voxel-grid approaches, Gaussian primitives offer a continuous density field that facilitates dense optimization while maintaining the rendering speeds necessary for closed-loop control. Recent frameworks such as [108] and [109] have demonstrated that leveraging 3D Gaussians enables superior convergence in camera pose estimation and map construction compared to implicit neural SLAM systems, primarily due to the efficiency of tile-based rasterization and the explicit nature of geometric updates. These systems exploit silhouette masks and adaptive expansion strategies to incrementally reconstruct scenes from unposed RGB-D streams, effectively handling the ambiguities inherent in dynamic settings where static assumptions fail.

A critical advancement in this domain is the transition from purely visual reconstruction to physically embodied world models that support safe navigation. The explicit parameterization of Gaussians allows for the direct formulation of collision constraints without the computational overhead of mesh extraction. [110] exemplifies this by formulating rigorous collision checks directly on Gaussian primitives to generate guaranteed-safe polytope corridors, enabling trajectory planning at 5 Hz and state estimation at 20 Hz—an order of magnitude faster than Neural Radiance Field (NeRF) based counterparts. Furthermore, the semantic capabilities of Gaussian Splatting are being harnessed to enhance robotic understanding; [111] and [96] integrate semantic features into the optimization loop, allowing robots to distinguish between static infrastructure and dynamic agents, thereby preventing erroneous map updates caused by moving objects. This semantic disentanglement is crucial for long-term autonomy in populated environments, as seen in [4], though Gaussian-based approaches offer significantly higher fidelity in novel view synthesis and geometric consistency.

Beyond navigation, the explicit nature of Gaussian primitives enables interactive manipulation and predictive simulation. [45] leverages editable Gaussian representations to create digital twins that update in real-time during manipulation tasks, facilitating open-vocabulary robotic interaction through semantic affordance distillation. Moreover, the fusion of visual rendering with physical dynamics addresses the limitation of purely photometric models. [26] introduces a dual particle-

Gaussian representation where visual forces derived from rendering errors correct physical states online, allowing robots to anticipate future scenarios and maintain synchronization with reality in dynamic worlds. Similarly, [107] extends these capabilities to low-light conditions by learning neural illumination models, ensuring robust exploration where traditional visual odometry typically degrades.

Despite these advancements, challenges remain in scaling these systems to large-scale outdoor environments and handling extreme motion blur. While [112] and [113] have made strides in modeling dynamic urban scenes by decomposing static backgrounds and moving vehicles, the reliance on accurate initial poses or external trackers limits full autonomy in unstructured settings. Future research must focus on developing globally optimized, pose-free frameworks that can jointly estimate camera trajectories and dynamic Gaussian attributes from sparse monocular inputs, as explored in [114] and [66]. Ultimately, the convergence of high-fidelity dynamic rendering, semantic understanding, and physical plausibility positions Gaussian Splatting as a foundational technology for the next generation of autonomous agents capable of operating safely and intelligently in complex, time-varying worlds.

6.2 Immersive Media, Streaming, and Human-Centric Applications

Bridging the gap between the physically embodied world models for robotics discussed previously and the benchmarking protocols detailed subsequently, the deployment of dynamic Gaussian Splatting in immersive media and human-centric applications necessitates a rigorous reconciliation between high-fidelity spatiotemporal reconstruction and the severe bandwidth constraints of real-time streaming. Unlike the static assumptions often sufficient for navigation, dynamic environments in telepresence and entertainment introduce a compounding storage overhead due to the temporal evolution of primitive attributes, demanding novel compression paradigms and adaptive transmission protocols. Recent advancements have shifted from naive per-frame optimization to compact, factorized representations that disentangle motion from appearance. For instance, [40] introduces an on-the-fly training framework utilizing a Neural Transformation Cache to model rigid transformations, significantly reducing the storage footprint required for free-viewpoint video streaming while maintaining interactive frame rates. Similarly, [46] proposes a decomposed neural radiance field strategy that categorizes spatiotemporal points into static, deforming, and emerging regions, enabling a hybrid feature streaming scheme that optimizes bandwidth allocation based on temporal characteristics.

In the domain of human performance capture, the explicit nature of Gaussian primitives facilitates unprecedented levels of controllability and anatomical fidelity, addressing the limitations of implicit methods that often struggle with the trade-off between training speed and rendering resolution for full-body avatars. [42] addresses this by leveraging the SMPL body model to initialize Gaussian distributions, allowing for the disentanglement of the human avatar from the background within minutes of training on monocular video. This approach enables novel-pose synthesis and animatability, critical for telepresence applications. Building upon skeletal priors, [115] employs cage deformations driven by joint angles rather than linear blend skinning, achieving superior visual quality for varied body shapes and motions. Furthermore, [24] integrates Gaussian splats directly onto a triangle mesh using barycentric coordinates, creating a hybrid representation where the mesh handles low-frequency motion and Gaussians capture high-frequency details, resulting in rendering speeds exceeding 300 FPS on modern GPUs.

To make these high-fidelity representations viable for volumetric video delivery, compression strategies must exploit the redundancy inherent in dense Gaussian clouds across time steps through advanced quantization and pruning techniques. [53] demonstrates that network pruning principles can be adapted to identify and remove insignificant Gaussians, while distilling spherical harmonics to lower degrees, achieving a 15x reduction in storage with minimal perceptual loss. In surgical contexts, where resource limitations are acute and mirror the constraints of edge robotics, [39] introduces deformation-aware pruning and 4D feature field condensation, reducing storage requirements by over 9x without compromising the real-time rendering efficiency necessary for robotic assistance. Additionally, [116] pioneers context-based compression by leveraging hash-grid structures to model spatial relations among anchors, yielding size reductions of over 75x compared to vanilla implementations.

Despite these strides, challenges persist in balancing extreme compression ratios with the preservation of high-frequency temporal details and view consistency, a tension that directly impacts the metrics used to evaluate such systems. Current streaming protocols often rely on progressive loading or level-of-detail strategies, yet the dynamic nature of human motion complicates static LOD hierarchies. Future research must focus on developing semantic-aware streaming mechanisms that prioritize visually salient regions, such as facial expressions and hand gestures, and integrate generative priors to hallucinate missing high-frequency details at the decoder side. These developments are essential not only for pushing the boundaries of bandwidth-efficient immersive experiences but also for establishing the robust performance baselines required by the comprehensive evaluation frameworks and datasets examined in the following section.

6.3 Benchmarking Datasets, Metrics, and Evaluation Protocols

The rigorous evaluation of dynamic Gaussian Splatting methodologies necessitates a critical examination of existing benchmarks, which have rapidly evolved from static object-centric collections to complex, large-scale spatiotemporal environments. Contemporary datasets now increasingly integrate multimodal sensors to provide ground truth for both photometric fidelity and geometric precision. For instance, the [117] dataset establishes a high-fidelity benchmark for human performance capture, offering 12MP multi-view footage essential for evaluating fine-grained deformation and texture consistency in articulated subjects. In the domain of autonomous driving, [49] and [113] leverage large-scale urban sequences, often incorporating LiDAR sweeps and semantic annotations to validate reconstruction scalability and dynamic object separation over geospatial footprints spanning hundreds of kilometers. Similarly, [112] and [94] utilize datasets like KITTI and Virtual KITTI 2 to assess surround-view synthesis and holistic scene understanding, emphasizing the need for benchmarks that capture complex occlusion relationships and varying illumination conditions. Specialized domains are also emerging, with [29] highlighting the scarcity of annotated endoscopic video data for deformable tissue reconstruction, while [118] provides a dedicated benchmark for high-framerate facial dynamics.

Despite the proliferation of datasets, the field faces a significant disconnect between standard image-based metrics and the actual quality of 4D reconstructions. Traditional metrics such as Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS) remain the de facto standards, as utilized in evaluations for [15], [7], and [16]. However, these metrics primarily assess pixel-level alignment and often fail to penalize geometric inconsistencies, temporal flickering, or “floaters” that do not significantly impact aggregate error scores. As noted in comparative studies like [119], high visual scores can mask severe structural inaccuracies, particularly in methods that prioritize rendering speed over geometric plausibility. This limitation is exacerbated in dynamic settings where motion blur and rolling shutter effects, addressed by [18] and [19], introduce ambiguities that static frame comparisons cannot resolve.

To address these deficiencies, emerging evaluation protocols must incorporate metrics specifically designed for temporal coherence and geometric stability. Future benchmarks should quantify trajectory smoothness, perhaps via optical flow consistency errors or deformation field regularity, to measure the physical plausibility of motion as advocated in [13] and [7]. Furthermore, geometric accuracy assessments should move beyond depth matching to include surface normal consistency and mesh extraction quality, leveraging techniques like those in [27] and [28] which enforce surface alignment. The [120] represents a step toward standardized protocols by introducing hidden test sets to prevent overfitting, a practice that should be extended to dynamic scenarios with sparse-view constraints as explored in [104] and [121]. Ultimately, a unified evaluation framework must balance rendering fidelity with geometric and temporal integrity, ensuring that advancements in dynamic Gaussian Splatting translate to robust performance in real-world applications ranging from robotic navigation to immersive media.

6.4 Computational Efficiency, Optimization, and Hardware Acceleration

Building upon the evaluation frameworks and metric limitations discussed previously, the computational viability of dynamic Gaussian Splatting hinges on resolving the tension between the massive parameter space of spatiotemporal primitives and the stringent latency requirements of real-time applications. While the explicit nature of Gaussian primitives offers inherent advantages over implicit volumetric methods, naive extensions to dynamic scenes often incur prohibitive memory costs and training overheads that undermine the efficiency-accuracy trade-offs highlighted in benchmark assessments. Recent algorithmic innovations have therefore prioritized memory-efficient representations and aggressive pruning strategies to address these gaps. For instance, [53] demonstrates that network pruning principles can be adapted to identify and remove insignificant Gaussians, achieving a 15x compression rate while boosting frame rates beyond 200 FPS. Similarly, [52] employs vector quantization and codebooks to compress geometric attributes and view-dependent colors, reducing storage by over 25x without compromising fidelity. In the dynamic domain, [39] extends these concepts to 4D primitives, utilizing deformation-aware pruning and feature field condensation to overcome the storage bottlenecks inherent in high-resolution surgical reconstructions.

Parallel to memory reduction, significant efforts have been directed toward accelerating training and eliminating dependencies on costly preprocessing pipelines, a critical step for improving the robustness noted as lacking in current evaluation protocols. Traditional Gaussian Splatting relies heavily on Structure-from-Motion (SfM) for initialization, a process that is often fragile in dynamic or texture-less environments. To address this, [66] integrates dense stereo priors from models like DUSt3R to enable pose-free, sparse-view reconstruction in under 40 seconds, effectively bypassing the SfM bottleneck. Complementary approaches, such as [122] and [123], propose optimization strategies that allow convergence from random initializations by leveraging frequency-domain analysis or distilling structure from coarse NeRF models. For dynamic sequences, [40] introduces a Neural Transformation Cache (NTC) to model frame-to-frame transformations, enabling on-the-fly training within 12 seconds per frame, a critical advancement for streaming applications where offline optimization is infeasible. Furthermore, [124] accelerates dynamic field optimization by utilizing time-aware voxel features and multi-distance interpolation, reducing training times to mere minutes while maintaining high rendering quality.

Hardware acceleration and rendering throughput are further enhanced through specialized rasterization techniques and Level-of-Detail (LOD) management, directly addressing the temporal flickering and geometric consistency issues raised in prior metric analyses. The standard sorting bottleneck in splatting, particularly for large-scale dynamic scenes, is mitigated by hierarchical approaches. [125] proposes a hierarchical rasterization method that systematically resorts and culls splats, eliminating view-dependent popping artifacts while doubling rendering performance through reduced primitive counts. For city-scale or unbounded environments, [126] and [127] employ divide-and-conquer training strategies coupled with LOD structures, ensuring consistent real-time performance across varying viewing distances by aggregating block-wise detail levels. Additionally, [128] combines radiance field priors with aggressive point pruning and test-time filtering to achieve rendering speeds exceeding 900 FPS, demonstrating that hybrid supervision can significantly robustify the optimization landscape against complex scene geometries.

Despite these advances, challenges remain in balancing extreme compression with the preservation of high-frequency temporal details and physical plausibility, reinforcing the need for the next-generation evaluation metrics proposed earlier. Emerging trends suggest a shift toward context-aware compression frameworks, such as [116], which leverages spatial consistencies between unorganized anchors and structured hash grids to achieve over 75x size reduction. Future research must further integrate these efficiency gains with physics-informed constraints and generative priors to enable scalable, interactive dynamic world models that operate seamlessly on edge devices without sacrificing the spatiotemporal coherence required for immersive and robotic applications.

7 Conclusion and Future Research Directions

The landscape of dynamic 3D scene reconstruction has undergone a paradigmatic shift with the advent of Gaussian Splatting, transitioning from slow, implicit volumetric methods to explicit, real-time representations capable of handling complex non-rigid motions. While current methodologies, such as 4D Gaussian Splatting [5] and canonical space warping techniques [7], have achieved remarkable fidelity and frame rates, significant challenges persist in extreme motion scenarios, topological changes, and unconstrained acquisition settings. A critical limitation remains the reliance on accurate camera poses and synchronized multi-view inputs, which restricts deployment to controlled environments. Although recent advances like InstantSplat [66] and BARF-inspired joint optimization [129] begin to address pose uncertainty, robust self-supervised reconstruction from in-the-wild, asynchronous internet videos remains an open frontier. Future research must prioritize the development of fully pose-free frameworks that leverage geometric priors from foundation models to disentangle camera motion from scene dynamics without explicit supervision.

Furthermore, the physical plausibility of dynamic reconstructions requires deeper integration with generative and neuro-symbolic intelligence. Current approaches often struggle with specular reflections and motion blur, artifacts that degrade visual fidelity in high-frequency dynamic regions. Techniques such as NeRF-DS [81] and BAD-Gaussians [18] offer preliminary solutions by modeling view-dependent effects and exposure trajectories, yet a unified framework that jointly optimizes geometry, appearance, and physical consistency is lacking. The future lies in coupling explicit Gaussian primitives with differentiable physics engines, as envisioned in physically embodied world models [26], to enforce conservation laws and collision constraints. This convergence will enable not only passive view synthesis but also interactive simulation and robotic planning in dynamic environments. Additionally, the integration of large-scale diffusion priors, as seen in DreamGaussian [130] and Vidu4D [131], holds promise for hallucinating occluded regions and synthesizing novel motion sequences, effectively transforming reconstruction into a generative task.

Finally, the scalability and editability of dynamic scenes necessitate adaptive architectures that seamlessly switch between static and dynamic representations based on detected motion patterns. Hybrid approaches like SC-GS [13] demonstrate the efficacy of decoupling motion bases from appearance, yet standardized evaluation protocols for long-term temporal consistency are absent. The community must move beyond frame-based photometric metrics like PSNR and SSIM, which often mask structural instabilities, toward metrics that quantify trajectory smoothness and causal consistency. By addressing these limitations through unified, physics-informed, and generative frameworks, the field can advance toward robust, generalizable dynamic neural rendering capable of operating in the unstructured complexity of the real world.

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[5] 4D Gaussian Splatting for Real-Time Dynamic Scene Rendering

[6] Real-time Photorealistic Dynamic Scene Representation and Rendering with 4D Gaussian Splatting

[7] Dynamic 3D Gaussians Tracking by Persistent Dynamic View Synthesis

[8] Deformable 3D Gaussians for High-Fidelity Monocular Dynamic Scene Reconstruction

[9] Temporally coherent 4D reconstruction of complex dynamic scenes

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